A-A-51314
26 DECEMBER 1985

COMMERCIAL ITEM DESCRIPTION

LIGHT, BED, HOSPITAL

The General Services Administration has authorized the use of this commercial item description, which replaces MIL-L-37502.

This commercial item description covers an adjustable bed light for use in medical treatment facilities.

Salient characteristic:

Shall be a universally-adjustable bed light suitable for use as a reading light and as a patient examining light. The light shall have a steel construction with an aluminum alloy reflector. The finish of the light shall be baked enamel. The light shall meet the requirements of UL-153 and UL-544. The light shall have a configuration similar to figure 1.

Sufficient slack shall be provided in the internal wiring so that no tension shall be developed as the assembly is articulated through any allowable adjustment. Wiring shall also be protected against rubbing or fraying, especially in the joints.

The base shall be constructed of steel. An on-off switch shall be mounted on the base or on the reflector. The light shall be constructed to mount in the intravenous socket of a hospital bed or in an auxiliary mounting bracket, NSN 6230-00-524-0403.

When fully extended, the outside of the reflector shall be at least 30 inches (76.2 cm) from the base axis. The light shall be constructed such that the extension arms of the light can rotate at least 60 degrees to either side of the horizontal plane. The extension arms shall consist of two rigid arms, made from steel tubing or rod joined by a vertically adjustable elbow joint. Either one or both arms may be of single or double rod construction. The elbow joint shall maintain any setting from fully retracted (with reflector touching the inner arm) to fully extended. The joint which connects the inner arm to the base shall allow the inner arm to retain any position from horizontal through vertical. After undergoing the stability test as specified, the light shall retain any position to which it is adjusted, with a maximum deviation of one inch (2.54 cm) in any direction. The joints shall retain their settings by means of balance mechanisms or compression-spring loaded friction joints. Thumb screws, wing nuts, or other methods of locking the adjustments shall not be employed.

FSC 6230

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

AMSC N/A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:54Z
Check the source to verify that this is the current version before use.

The reflector shall be made of aluminum at least 0.03 (.076 cm) inch thick. Movement of the reflector about the outer extension arm shall be as shown in figures 2a, 2b, and 2c. The reflector shall be of a double shell construction. The reflector shall also be vented to prevent overheating when the open end of the reflector is blocked. The lamp shall be recessed within the reflector at least 3/4 inch (1.91 cm) or the open end of the reflector shall have a suitable guard. The exterior of the reflector shall not exceed 122°F (50°C) at any point during the normal operation of the light. When tested for fire hazard as specified the light shall not cause the temperature of the thermocouple to exceed 149°C (300°F).

The bed light shall be designed to accommodate, and be provided with, a medium screw base, 60 watt, 115 volt, AC incandescent lamp. The lamp shall provide at least 30 foot-candles of illumination at a distance of two feet from the outer edge of the reflector.

The light shall be furnished with a nominal 9 feet (2.74 m) of three conductor cord (with ground lead) fitted with a Hospital Grade, grounded, three-blade plug as specified in UL544, which is compatible with the receptacle specified in American National Standards Institute (ANSI) Standard C73.11.

No electrical breakdown shall occur when the light is tested for dielectric strength as specified.

The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of Underwriters' Laboratories (UL), Inc., Standard for safety UL153 and the applicable portions of UL544. Proof shall consist of the following:

A. UL label affixed to the product, provided the contractor agrees to have Underwriters' Laboratories, Inc., forward a copy of the UL report directly to the contracting officer, upon the latter's request.

B. Laboratory report from a nationally recognized independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The Government reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed finding of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to ensure identity of production items with test information, such as material used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts list, all appropriately identified. Compliance with this paragraph does not excuse compliance with all other requirements of the contract and specifications.

Workmanship. The bed lights shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Each (EA). One bed light, as specified, constitutes one unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:54Z
Check the source to verify that this is the current version before use.

Quality Assurance Provision.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by The Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Testing shall include but not be limited to the following:

Stability. The light shall be fully retracted, with the reflector, or the reflector, in contact with the inner extension arm and the reflector pointing straight down. The arm shall then be fully extended. The reflector shall be rotated to the full up position, and also tilted (at the reflector joint) to the full up position. The light shall then be returned to its original configuration. This test shall be repeated for 50 complete cycles. The light shall then be examined for distortion and for failure to retain any setting. The internal wiring shall be examined for evidence of strain or damage to the insulation.

Dielectric strength. The light shall be tested in accordance with section 27 of UL544.

Polarity. A continuity check shall be made to determine if the switch is wired directly to the ungrounded conductor terminal on the plug as specified in UL153.

Temperature. The light, equipped with a 60 watt bulb, shall be left on for at least three hours with the illumination directed straight down. After three hours, the temperature of the outside of the reflector shall be measured with a thermocouple or an optical pyrometer with an accuracy of +/- 5 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:54Z
Check the source to verify that this is the current version before use.

Fire hazard. The light, equipped with a 60 watt bulb, shall be adjusted so that the illumination is straight down. The light shall be turned on and the front of the reflector pushed, with at least 10 pounds of force, into direct contact with a cotton bedsheet placed on a firm mattress or pillow. (The pressure need not be maintained after initial application). A thermocouple shall be placed under the sheet within the circumference of the reflector. The light shall be left on for one hour, and the temperature shall be recorded at the end of the hour.

Leakage current. The light shall be tested in accordance with Section 27 of UL544.

Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table I. Acceptance level shall be zero for all sample sizes.

TABLE I. Sampling criteria for tests.

Characteristic	Inspection level	AQL (defects per hundred units)
Stability	S-1	3.0
Dielectric strength	II	1.0
Polarity	II	1.0
Temperature	S-1	1.0
Fire Hazard	S-1	1.0

Examination. The light shall be examined to compliance with specified requirements. Examination shall be conducted in accordance with table II.

TABLE II. Classification of defects.

Categories	Defects
Major	
101	Component part missing.
102	Any adjustable part fails to move easily.
103	Component parts in assembled light cracked, torn or ruptured.
104	Enameled surfaces have unenameled or corroded areas.
105	Ground wire not adequately secured to frame or enclosure.
106	Interior of wire raceway, including joints, has burrs or sharp edges.
107	Wire stain relief inadequate in any position of adjustment.
108	Wire is strained at any joint, in the reflector, or at any point of articulation.
109	Ungrounded conductor has insulation colored white, natural gray or green.
Minor	
201	Assembled light exhibits dirt, oil, grease, chips or foreign matter.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:54Z
Check the source to verify that this is the current version before use.

TABLE III. Sampling for examination.

Defect type	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	4.0

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

6230-00-145-1407

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:54Z
Check the source to verify that this is the current version before use.

A-A-51314

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATION ACTIVITY:

VA-OSS

PGC 41191

Project No. 6230-0290

Location: NCR/CID-LIGHT-BED-HOSPITAL

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:54Z
Check the source to verify that this is the current version before use.

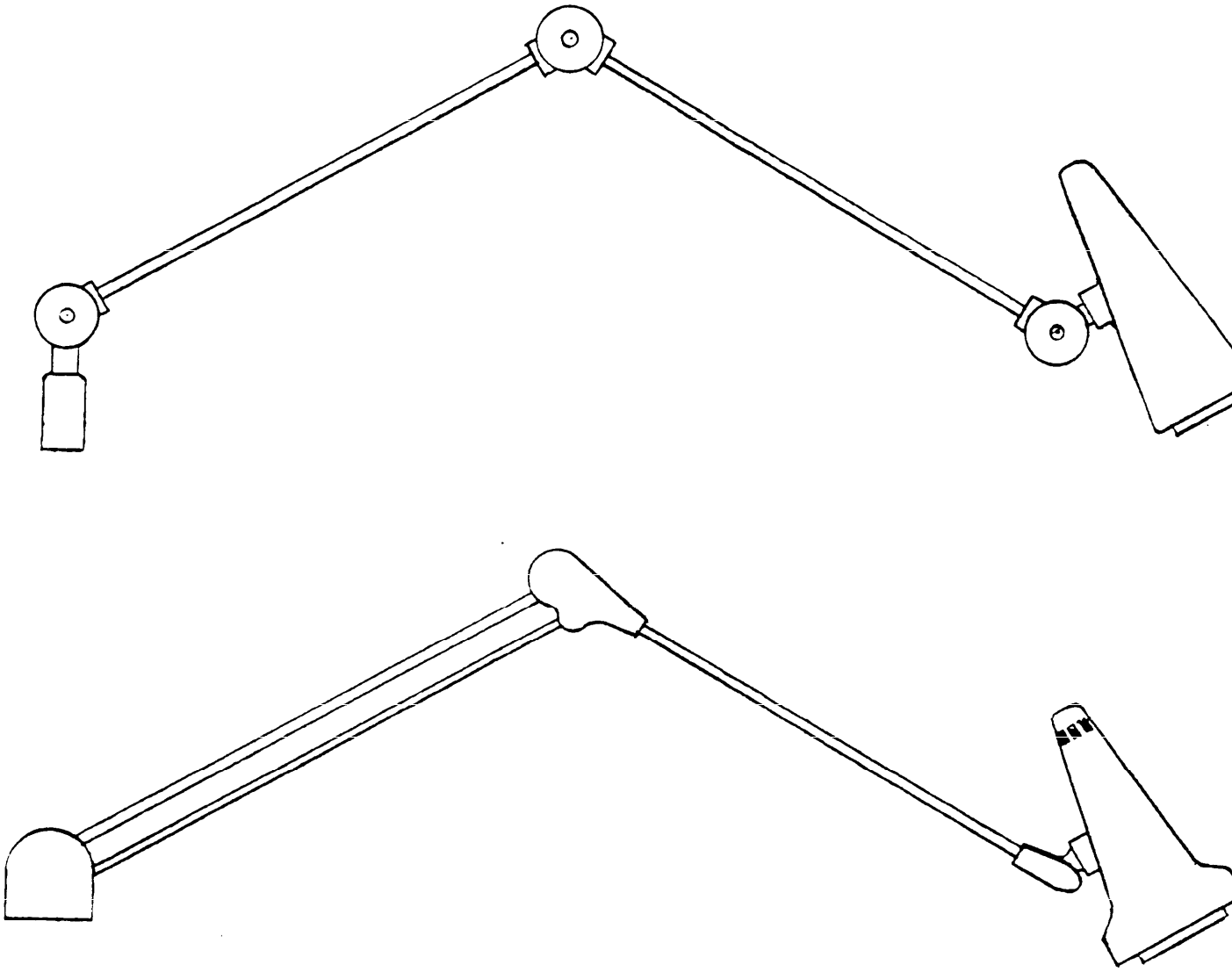
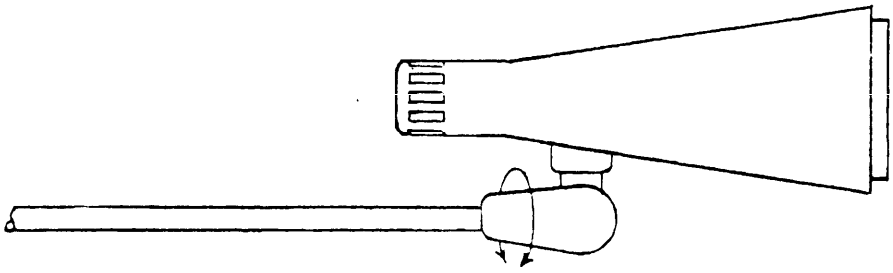


FIGURE I. Light, bed. Typical configurations.

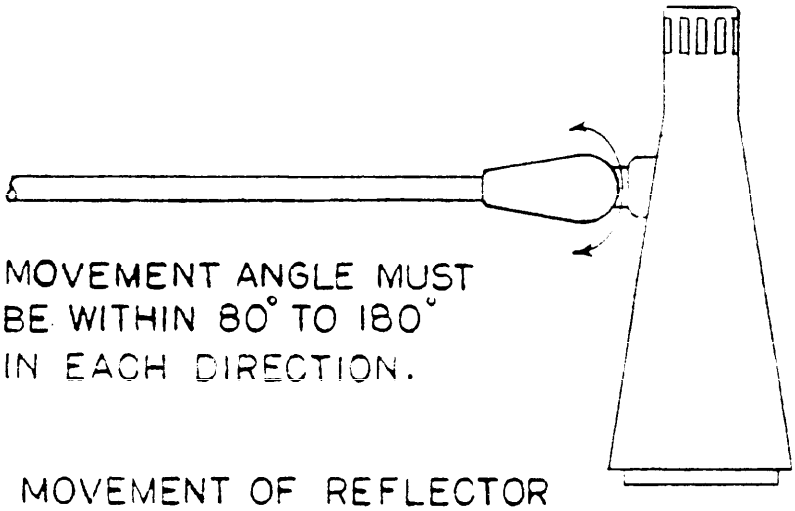
NOTE: FIGURES ARE FOR ILLUSTRATIVE PURPOSES AND ARE NOT TO BE USED AS SPECIFIED REQUIREMENTS.

Source: https://assist.dla.mil -- Downloaded: 202610-26T08:54Z
Check the source to verify that this is the currentversion before use.



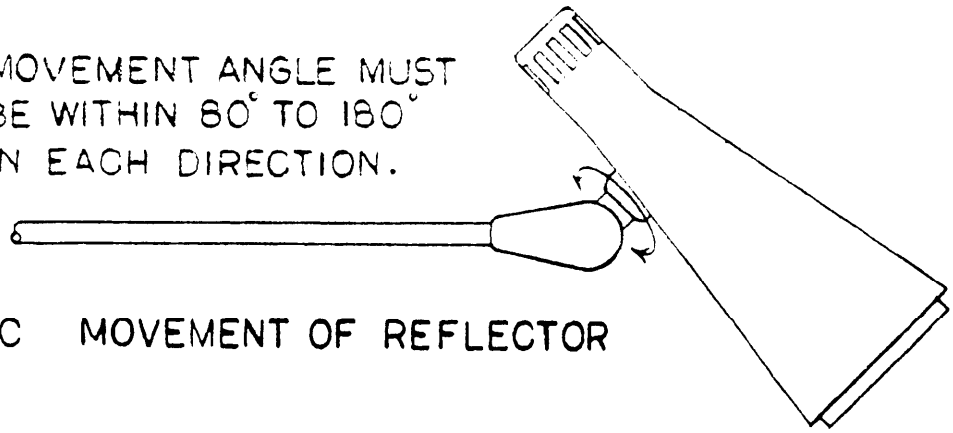
MOVEMENT ANGLE MUST
BE WITHIN 80° TO 180°
IN EACH DIRECTION.

FIGURE 2A - MOVEMENT OF REFLECTOR



MOVEMENT ANGLE MUST
BE WITHIN 80° TO 180°
IN EACH DIRECTION.

FIGURE 2B MOVEMENT OF REFLECTOR



MOVEMENT ANGLE MUST
BE WITHIN 80° TO 180°
IN EACH DIRECTION.

FIGURE 2C MOVEMENT OF REFLECTOR

NOTE: FIGURES REPRESENT MOVEMENT
OF REFLECTOR ONLY, THE SIZE
AND SHAPE ARE ILLUSTRATIVE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:54Z
Check the source to verify that this is the current version before use.